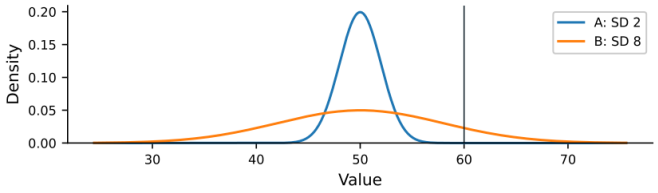
MSC 201 **92847**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A U(0,10) sample has mean 5.4. Must the model be wrong?	A. Yes; all uniform values must equal 5. B. No; a sample mean can differ from the model mean. C. No; any mean proves the model. D. Yes; every sample must average exactly 5. 10 pts
2. A uniform wait ranges from 0 to 10. Only 10% exceed a cutoff. Where is that cutoff?	A. At 5. B. At 10. C. At 9. D. At 1. 10 pts
3. For an exponential model, P(X>3)=0.40. Given X>7, what is the chance of more than 3 additional minutes?	A. 0. B. 0.40 divided by 7. C. 0.40. D. 1. 10 pts
4. Normal models A and B share mean 50, with SDs 2 and 8. Which has more probability above 60? 	A. A. B. They have equal upper tails. C. B. D. Both probabilities are exactly zero. 10 pts
5. A normal model's mean increases, with SD fixed. At a fixed deadline, what happens to the probability of finishing before it?	A. It becomes exactly 0.5. B. It decreases. C. It stays unchanged. D. It increases. 10 pts
6. P(X>2)=0.50 and P(X>6)=0.125 for an exponential wait. What is P(2<X<6)?	A. 0.125. B. 0.50. C. 0.375. D. 0.625. 10 pts

Test continues on the other side of the sheet.

MSC 201 **92847**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A proposed density dips below zero over an interval. What is wrong?	A. It would assign negative area to that interval. B. Its maximum must equal 1. C. A density must be flat everywhere. D. Every valid density must be symmetric. 10 pts
8. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They all become positive. B. They stay unchanged. C. They multiply by 100. D. They divide by 100. 10 pts
9. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 45%. B. 5%. C. 90%. D. 10%. 10 pts
10. About 95% of a normal model lies within 2 SD. About what percent lies above mean+2 SD?	A. 5%. B. 95%. C. 2.5%. D. 47.5%. 10 pts

92847

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6.

C.

10 pts

TEST PAGE 2

7. A.

10 pts

8.

B.

10 pts

9.

B.

10 pts

10.

C.

10 pts

MSC 201 **49716**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Under an exact normal model, how do probabilities below 480 and at most 480 compare?	A. At most is larger by the density height. B. They differ by 1/480. C. Below is always zero. D. They are equal. 10 pts
2. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. A constant rate across the whole period. B. Allowing a long wait. C. Recording waits in minutes. D. Nonnegative waiting times. 10 pts
3. A rate is stated per hour but a cutoff is inserted as a number of minutes. What must be done before using the tail formula?	A. Express rate and cutoff in compatible time units. B. Replace the cutoff with the mean. C. Round the probability to an integer. D. Multiply both numbers by the sample size. 10 pts
4. A proposed normal model gives substantial probability to negative durations. What is the concern?	A. Its mean must be zero. B. Every duration must be negative. C. Normal distributions cannot model any positive variable. D. It assigns appreciable probability to impossible values. 10 pts
5. Someone reports F(4)=0.6 and F(8)=0.4 for one variable. Why is this impossible?	A. Cumulative functions must always equal 0.5. B. Every cutoff must have the same probability. C. Both numbers are below 1. D. A cumulative probability cannot decrease as the cutoff rises. 10 pts
6. X has mean 50, SD 10. Which standard-normal event corresponds to X<60?	A. Z<6. B. Z<-1. C. Z<1. D. Z>1. 10 pts
7. A continuous uniform model runs from 4 to 12. What is its mean?	A. 12. B. 8. C. 4. D. 16. 10 pts

Test continues on the other side of the sheet.

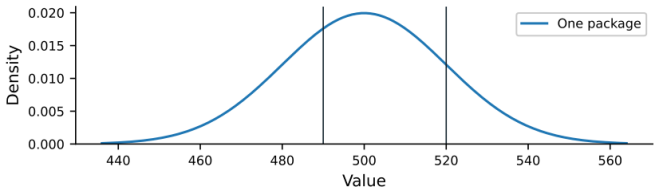
MSC 201 **49716**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Mean is 100, SD is 10, and the supplied 90th-percentile z is 1.28. What is the corresponding value?	A. 87.2. B. 112.8. C. 101.28. D. 12.8. 10 pts
9. A package is accepted if 490<mass<520. Which part of its normal curve should be shaded? 	A. Everything above 520. B. Only the point 500. C. Both outside tails. D. Only the area between 490 and 520. 10 pts
10. A U(0,12) wait has already lasted 11 minutes. What is the largest possible additional wait under this model?	A. There is no upper bound. B. 1 minute. C. 11 minutes. D. 12 minutes. 10 pts

49716

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6.

C.

10 pts

7.

B.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

D.

10 pts

10.

B.

10 pts

MSC 201 **16300**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Given that a U(0,12) wait exceeds 4, what shape describes the total wait's conditional density?	A. Flat over 4 to 12. B. Normal over all real numbers. C. A point at 4. D. Flat over 0 to 12 unchanged. 10 pts
2. X is uniform from 0 to 10. Which event is more likely?	A. They are equally likely. B. X>6. C. X=6. D. X<6. 10 pts
3. About 95% of a normal model lies within 2 SD. About what percent lies above mean+2 SD?	A. 47.5%. B. 2.5%. C. 5%. D. 95%. 10 pts
4. An exponential waiting time has mean 5 minutes. What rate per minute matches it?	A. 25. B. 0.2. C. 0.5. D. 5. 10 pts
5. A continuous model gives F(6)=0.70. What does that say?	A. 70% of modeled durations are at most 6. B. The mean is 6. C. 70% of durations equal exactly 6. D. The density height at 6 is 0.70. 10 pts
6. A probability interval is widened to include its original interval plus a region of positive density. What happens to its probability?	A. It increases. B. It exceeds 1 automatically. C. It decreases. D. It stays unchanged. 10 pts
7. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They stay unchanged. B. They divide by 100. C. They multiply by 100. D. They all become positive. 10 pts

Test continues on the other side of the sheet.

MSC 201 **16300**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A normal model's mean rises from 40 to 50 while SD stays 6. What happens to the curve?	A. It becomes wider. B. It becomes right-skewed. C. It shifts right without changing width. D. Its total area increases. 10 pts
9. For a normal distribution, how do Q1 and Q3 lie relative to the mean?	A. Q3 must be twice Q1. B. Both lie above it. C. They are equally distant on opposite sides. D. Both equal it. 10 pts
10. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. Nonnegative waiting times. B. Allowing a long wait. C. A constant rate across the whole period. D. Recording waits in minutes. 10 pts

16300

TEST PAGE 1

1. A.

10 pts

2. D.

10 pts

3. B.

10 pts

4. B.

10 pts

5. A.

10 pts

6. A.

10 pts

7. A.

10 pts

TEST PAGE 2

8. C.

10 pts

9. C.

10 pts

10. C.

10 pts

MSC 201 **86437**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A uniform wait ranges from 0 to 10. Only 10% exceed a cutoff. Where is that cutoff?	A. At 1. B. At 9. C. At 5. D. At 10. 10 pts
2. The course uses N(mean, SD). A technician expects readings from N(40,6) to cluster near 6. What is the best correction?	A. The readings cluster near 40; 6 describes their spread. B. Every reading must equal 40. C. The model allows only the two values 6 and 40. D. They cluster near 6; 40 is the sample size. 10 pts
3. A uniform interval centered at 10 doubles its width while keeping center 10. What happens?	A. Mean doubles; SD stays unchanged. B. Both double. C. Both stay unchanged. D. Mean stays 10; SD doubles. 10 pts
4. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It stays unchanged. B. It rises by 3. C. It rises by 90%. D. It falls by 3. 10 pts
5. A supplied normal table gives $P(Z<1.2)=0.8849$. What is $P(Z>1.2)$?	A. 0.1151. B. 1.8849. C. 0.44245. D. 0.8849. 10 pts
6. A normal variable has mean 40 and SD 5. What z-score corresponds to 50?	A. 2. B. -2. C. 10. D. 0.2. 10 pts
7. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. Yes; both are measured in grams. B. Yes; averages always vary more. C. No; averages have no units. D. No; the average has less spread. 10 pts

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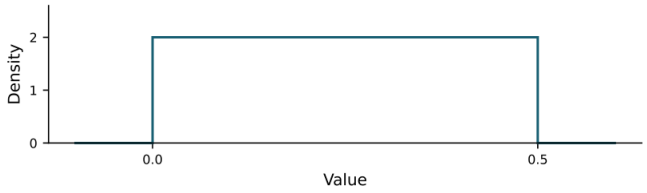
MSC 201 **86437**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. No; its total area is 1, and height is not probability. B. Yes; every density height must be at most 1. C. No; total area may be any positive number. D. Yes; its probability is 2 everywhere. 10 pts
9. An exponential model has mean wait 4 minutes. What is its SD?	A. 2 minutes. B. 16 minutes. C. 4 minutes. D. 0 minutes. 10 pts
10. An exponential wait has already lasted 10 minutes. Compared with a fresh wait, what is the distribution of the additional wait?	A. It becomes uniform. B. Its expected remaining duration is the original mean minus 10. C. It must end immediately. D. The same under the memoryless model. 10 pts

86437

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5. A.

10 pts

6. A.

10 pts

7.

D.

10 pts

TEST PAGE 2

8. A.

10 pts

9.

C.

10 pts

10.

D.

10 pts

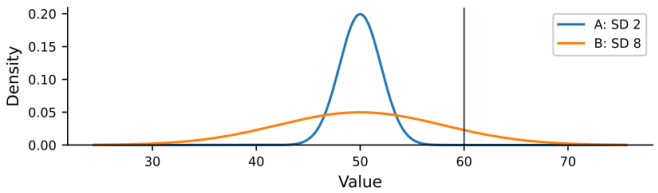
MSC 201 **57773**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. Both endpoints of a uniform model increase by 4. What happens to its mean and SD?	A. Mean rises by 4; SD stays unchanged. B. Both stay unchanged. C. Both rise by 4. D. SD rises by 4; mean stays unchanged. 10 pts
2. A proposed density dips below zero over an interval. What is wrong?	A. Its maximum must equal 1. B. It would assign negative area to that interval. C. A density must be flat everywhere. D. Every valid density must be symmetric. 10 pts
3. For a normal distribution, how do Q1 and Q3 lie relative to the mean?	A. Both equal it. B. They are equally distant on opposite sides. C. Both lie above it. D. Q3 must be twice Q1. 10 pts
4. For an exponential distribution with mean 5 minutes, how does its median compare?	A. The median exceeds 5. B. The median equals 5. C. The median is below 5. D. The median does not exist. 10 pts
5. An exponential wait has already lasted 10 minutes. Compared with a fresh wait, what is the distribution of the additional wait?	A. It must end immediately. B. It becomes uniform. C. The same under the memoryless model. D. Its expected remaining duration is the original mean minus 10. 10 pts
6. Normal models A and B share mean 50, with SDs 2 and 8. Which has more probability above 60? 	A. B. B. A. C. Both probabilities are exactly zero. D. They have equal upper tails. 10 pts

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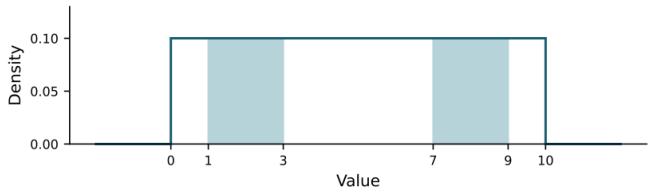
MSC 201 **57773**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Which model assigns equal probability to all equal-length subintervals within its support?	A. An exponential model. B. Every continuous model. C. A continuous uniform model. D. A normal model. 10 pts
8. About 95% of a normal model lies within 2 SD. About what percent lies above mean+2 SD?	A. 95%. B. 2.5%. C. 47.5%. D. 5%. 10 pts
9. Mean is 30 and SD is 4. Which value has z=1.5?	A. 31.5. B. 36. C. 26. D. 45. 10 pts
10. X is uniform from 0 to 10. Compare intervals 1 to 3 and 7 to 9. 	A. Their probabilities depend on the sample median. B. The second is more likely because its values are larger. C. They have equal probability. D. The first is more likely because it is closer to zero. 10 pts

57773

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. B.

10 pts

4. C.

10 pts

5. C.

10 pts

6. A.

10 pts

TEST PAGE 2

7. C.

10 pts

8. B.

10 pts

9. B.

10 pts

10. C.

10 pts

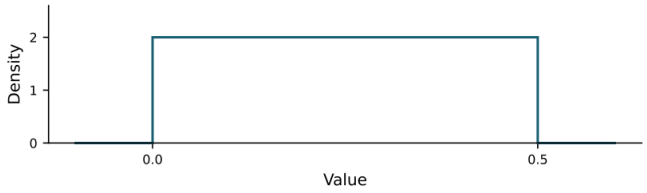
MSC 201 **81340**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X is uniform from 0 to 10. Which event is more likely?	A. X=6. B. X>6. C. They are equally likely. D. X<6. 10 pts
2. A question asks for the mass exceeded by only 10% of packages. What units should the answer have?	A. Probability units only. B. A percentage of sample size. C. Mass units. D. No units because it is a z-score automatically. 10 pts
3. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 0.15%. B. 0.3%. C. 3%. D. 99.7%. 10 pts
4. A density is flat at height 2 over a half-unit interval and zero elsewhere. Is height 2 alone a problem? 	A. No; total area may be any positive number. B. Yes; its probability is 2 everywhere. C. No; its total area is 1, and height is not probability. D. Yes; every density height must be at most 1. 10 pts
5. A rate is stated per hour but a cutoff is inserted as a number of minutes. What must be done before using the tail formula?	A. Multiply both numbers by the sample size. B. Round the probability to an integer. C. Replace the cutoff with the mean. D. Express rate and cutoff in compatible time units. 10 pts
6. One model is symmetric and bell-shaped; another has a long right tail and decreases from zero onward. Which pairing fits those shapes?	A. Exponential first; normal second. B. Normal first; exponential second. C. Both must be uniform. D. Uniform first; normal second. 10 pts

Test continues on the other side of the sheet.

MSC 201 **81340**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A normal model has mean 20. What is P(X<20)?	A. The SD. B. 0.5. C. 0. D. 1. 10 pts
8. Observed waits cluster near 0 and near 12, with few in the middle. What does that suggest about U(0,12)?	A. Its flat-density assumption deserves checking. B. A mean of 6 guarantees the uniform model. C. The range alone establishes the density. D. The uniform model is automatically proved. 10 pts
9. Mean is 30 and SD is 4. Which value has z=1.5?	A. 36. B. 26. C. 31.5. D. 45. 10 pts
10. A help desk records both the number of requests received in an hour and the exact time until its next request. Which is a continuous variable that an exponential model could describe?	A. Whether there was a request in the hour. B. The exact time until the next request. C. The number of requests received in the hour. D. The number of help desks open. 10 pts

81340

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

B.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

B.

10 pts

TEST PAGE 2

7.

B.

10 pts

8. A.

10 pts

9. A.

10 pts

10.

B.

10 pts

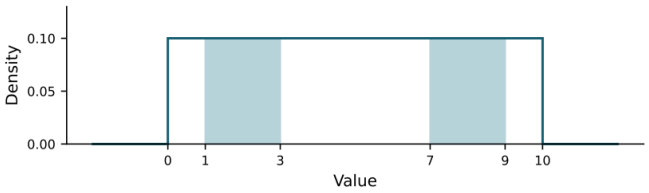
MSC 201 **74439**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A U(0,12) wait has already lasted 11 minutes. What is the largest possible additional wait under this model?	A. 12 minutes. B. 1 minute. C. There is no upper bound. D. 11 minutes. 10 pts
2. X is uniform from 0 to 10. Compare intervals 1 to 3 and 7 to 9. 	A. They have equal probability. B. Their probabilities depend on the sample median. C. The second is more likely because its values are larger. D. The first is more likely because it is closer to zero. 10 pts
3. An exact continuous duration has P(X=4)=0. Can a timer report 4 seconds?	A. Only if the density equals zero everywhere. B. No; 4 seconds can never be recorded. C. Yes; rounding collects a positive-width interval of exact values. D. Only if the model becomes discrete before rounding. 10 pts
4. Mean is 30 and SD is 4. Which value has z=1.5?	A. 45. B. 31.5. C. 36. D. 26. 10 pts
5. The mean lifetime is 8 hours. What does an expected total lifetime of 80 hours for ten items mean?	A. Every item lasts exactly 8 hours. B. Every set totals exactly 80 hours. C. All items fail simultaneously at hour 80. D. It is an average total over repeated sets, not a guaranteed total. 10 pts
6. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 90%. B. 45%. C. 10%. D. 5%. 10 pts

Test continues on the other side of the sheet.

MSC 201 **74439**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Normal models share SD 4 but have means 10 and 14. Which has more probability below 8?	A. The mean-14 model. B. They have equal probabilities. C. Both have probability 0.5. D. The mean-10 model. 10 pts
8. A supplied normal table gives P(Z<1.2)=0.8849. What is P(Z>1.2)?	A. 0.8849. B. 0.1151. C. 0.44245. D. 1.8849. 10 pts
9. The wait until the next request is modeled as exponential with rate 4 per hour. What is its mean in minutes?	A. 15 minutes. B. 4 minutes. C. 240 minutes. D. 60 minutes. 10 pts
10. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. Yes; both are measured in grams. B. Yes; averages always vary more. C. No; the average has less spread. D. No; averages have no units. 10 pts

74439

TEST PAGE 1

1.

B.

10 pts

2. A.

10 pts

3.

C.

10 pts

4.

C.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

B.

10 pts

9. A.

10 pts

10.

C.

10 pts

MSC 201 **77462**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A normal model has mean 20. What is P(X<20)?	A. 0. B. The SD. C. 1. D. 0.5. 10 pts
2. A U(0,12) wait has already lasted 11 minutes. What is the largest possible additional wait under this model?	A. 1 minute. B. There is no upper bound. C. 12 minutes. D. 11 minutes. 10 pts
3. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. The one with more recorded observations regardless of SD. B. The model with the smaller SD. C. They must have equal tails. D. The model with the larger SD. 10 pts
4. A nonnormal variable has a well-defined mean and SD. Can its z-score automatically be looked up as a standard-normal probability?	A. Yes; a positive SD is sufficient. B. Yes; z-scores always imply normality. C. No; only normal variables can be standardized. D. No; the normal shape assumption still matters. 10 pts
5. An exponential model has mean wait 4 minutes. What is its SD?	A. 16 minutes. B. 4 minutes. C. 2 minutes. D. 0 minutes. 10 pts
6. For a continuous duration, how do P(X<6) and P(X<=6) compare?	A. The first is always zero. B. The second is always twice the first. C. They differ by the density height at 6. D. They are equal. 10 pts

Test continues on the other side of the sheet.

MSC 201 **77462**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. A help desk's exponential waiting model is memoryless. Which prediction is justified after a request-free stretch?	A. The elapsed time so far must be added to the model's mean remaining wait. B. A request must arrive before the original mean wait expires. C. A request becomes more likely in the next minute merely because the wait has been long. D. For a fixed additional interval, the chance of no request matches the chance from a fresh start. 10 pts
8. Mean is 100, SD is 10, and the supplied 90th-percentile z is 1.28. What is the corresponding value?	A. 87.2. B. 101.28. C. 112.8. D. 12.8. 10 pts
9. X is uniform from 2 to 10. To find $P(0<X<5)$, which portion contributes area?	A. 0 to 2. B. 5 to 10. C. 0 to 5. D. 2 to 5. 10 pts
10. A supplied normal table gives $P(Z<1.2)=0.8849$. What is $P(Z>1.2)$?	A. 0.8849. B. 0.44245. C. 0.1151. D. 1.8849. 10 pts

77462

TEST PAGE 1

1.

D.

10 pts

2. A.

10 pts

3.

D.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

D.

10 pts

8.

C.

10 pts

9.

D.

10 pts

10.

C.

10 pts

MSC 201 **59640**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A normal model has mean 50 and SD 2. Which interval contains about 99.7%?	A. 46 to 54. B. 44 to 56. C. 48 to 52. D. 50 to 56. 10 pts
2. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 90%. B. 5%. C. 10%. D. 45%. 10 pts
3. A normal model's mean rises from 40 to 50 while SD stays 6. What happens to the curve?	A. Its total area increases. B. It becomes wider. C. It becomes right-skewed. D. It shifts right without changing width. 10 pts
4. For a normal variable, a cutoff above the mean is reported to have upper-tail probability 0.80. What should be checked?	A. The tail direction; that upper tail must be below 0.5. B. Whether an upper tail must be negative. C. Whether every normal curve has mean zero. D. Whether probabilities can exceed 0.5 at all. 10 pts
5. Compare one U(0,12) wait with the average of 100 independent such waits. Which is less variable?	A. The individual wait. B. The average must have zero spread. C. They have identical spread. D. The average. 10 pts
6. Waits are nonnegative. Is that sufficient to establish an exponential model?	A. Yes; every nonnegative variable is exponential. B. Yes; nonnegative variables have equal mean and SD. C. No; exponential values must be negative. D. No; the arrival process and distributional assumptions still need support. 10 pts
7. X has mean 50, SD 10. Which standard-normal event corresponds to X<60?	A. Z<1. B. Z>1. C. Z<-1. D. Z<6. 10 pts

Test continues on the other side of the sheet.

MSC 201 **59640**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. An exact continuous duration has P(X=4)=0. Can a timer report 4 seconds?	A. Only if the density equals zero everywhere. B. Only if the model becomes discrete before rounding. C. No; 4 seconds can never be recorded. D. Yes; rounding collects a positive-width interval of exact values. 10 pts
9. After an 11-minute wait, U(0,12) allows at most 1 more minute. An exponential model allows longer additional waits. What explains the difference?	A. Both models are memoryless. B. Exponential waits stop at their mean. C. Uniform waits have a finite endpoint; exponential waits do not. D. Uniform waits are always zero. 10 pts
10. Uniform model A spans width 4; model B spans width 8. Which density is taller?	A. A. B. B. C. The one with the larger mean regardless of width. D. They have equal height. 10 pts

59640

TEST PAGE 1

1. B.

10 pts

2. B.

10 pts

3. D.

10 pts

4. A.

10 pts

5. D.

10 pts

6. D.

10 pts

7. A.

10 pts

TEST PAGE 2

8. D.

10 pts

9. C.

10 pts

10. A.

10 pts

MSC 201 **35444**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. P(X>2)=0.50 and P(X>6)=0.125 for an exponential wait. What is P(2<X<6)?	A. 0.375. B. 0.50. C. 0.625. D. 0.125. 10 pts
2. An exact continuous duration has P(X=4)=0. Can a timer report 4 seconds?	A. Yes; rounding collects a positive-width interval of exact values. B. Only if the model becomes discrete before rounding. C. Only if the density equals zero everywhere. D. No; 4 seconds can never be recorded. 10 pts
3. Two normal curves share SD but have different means. What is equal?	A. Their widths and shapes. B. Their centers. C. Their probabilities below every fixed cutoff. D. Their 90th-percentile values. 10 pts
4. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. The second is larger because its mean is larger. B. They are equal. C. Both standard deviations are zero. D. The first is larger because it includes zero. 10 pts
5. For a normal variable, a cutoff above the mean is reported to have upper-tail probability 0.80. What should be checked?	A. Whether probabilities can exceed 0.5 at all. B. The tail direction; that upper tail must be below 0.5. C. Whether every normal curve has mean zero. D. Whether an upper tail must be negative. 10 pts
6. The mean lifetime is 8 hours. What does an expected total lifetime of 80 hours for ten items mean?	A. Every set totals exactly 80 hours. B. Every item lasts exactly 8 hours. C. All items fail simultaneously at hour 80. D. It is an average total over repeated sets, not a guaranteed total. 10 pts

Test continues on the other side of the sheet.

MSC 201 **35444**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Only 5% exceed cutoff k. Which percentile is k?	A. 100th. B. 5th. C. 50th. D. 95th. 10 pts
8. A value equals its distribution's mean. What is its z-score?	A. The mean. B. The SD. C. 1. D. 0. 10 pts
9. X is uniform from 0 to 10. Which event is more likely?	A. X=6. B. They are equally likely. C. X<6. D. X>6. 10 pts
10. A normal model's mean increases, with SD fixed. At a fixed deadline, what happens to the probability of finishing before it?	A. It increases. B. It becomes exactly 0.5. C. It decreases. D. It stays unchanged. 10 pts

35444

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. A.

10 pts

4. B.

10 pts

5. B.

10 pts

6. D.

10 pts

TEST PAGE 2

7. D.

10 pts

8. D.

10 pts

9. C.

10 pts

10. C.

10 pts

MSC 201 **56381**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. No; averages have no units. B. No; the average has less spread. C. Yes; both are measured in grams. D. Yes; averages always vary more. 10 pts
2. For a normal distribution, how do Q1 and Q3 lie relative to the mean?	A. Both lie above it. B. They are equally distant on opposite sides. C. Both equal it. D. Q3 must be twice Q1. 10 pts
3. A model predicts 5% of items fall outside specifications. Among 200 items, how many are expected outside?	A. Exactly 10 must occur. B. 40. C. 10. D. 5. 10 pts
4. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They all become positive. B. They multiply by 100. C. They divide by 100. D. They stay unchanged. 10 pts
5. $F(3)=0.20$ and $F(7)=0.80$ for a continuous variable. What is $P(3<X<7)$?	A. 0.80. B. 1.00. C. 0.20. D. 0.60. 10 pts
6. Waits are nonnegative. Is that sufficient to establish an exponential model?	A. No; the arrival process and distributional assumptions still need support. B. Yes; nonnegative variables have equal mean and SD. C. No; exponential values must be negative. D. Yes; every nonnegative variable is exponential. 10 pts
7. An exponential arrival rate doubles. What happens to mean waiting time?	A. It halves. B. It doubles. C. It stays unchanged. D. It becomes zero. 10 pts

Test continues on the other side of the sheet.

MSC 201 **56381**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. X is uniform from 0 to 12. Given X>4, what is P(X>8)?	A. 4/12. B. 8/12. C. 8/8. D. 4/8. 10 pts
9. A wait is uniform from 0 to 20 minutes. What is its 75th percentile?	A. 15 minutes. B. 20 minutes. C. 75 minutes. D. 5 minutes. 10 pts
10. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. The center shifts right. B. It becomes higher and narrower. C. The total area rises above 1. D. It becomes lower as the curve spreads out. 10 pts

56381

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3.

C.

10 pts

4.

D.

10 pts

5.

D.

10 pts

6. A.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

D.

10 pts

9. A.

10 pts

10.

D.

10 pts

MSC 201 **82278**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A proposed density dips below zero over an interval. What is wrong?	A. It would assign negative area to that interval. B. Its maximum must equal 1. C. Every valid density must be symmetric. D. A density must be flat everywhere. 10 pts
2. A supplied positive critical z leaves 0.025 to its right. What cumulative area lies to its left?	A. 0.025. B. 0.975. C. 0.95. D. 0.5. 10 pts
3. An exponential model has mean wait 4 minutes. What is its SD?	A. 16 minutes. B. 0 minutes. C. 4 minutes. D. 2 minutes. 10 pts
4. Two normal curves share SD but have different means. What is equal?	A. Their centers. B. Their widths and shapes. C. Their 90th-percentile values. D. Their probabilities below every fixed cutoff. 10 pts
5. An exponential wait has already lasted 10 minutes. Compared with a fresh wait, what is the distribution of the additional wait?	A. The same under the memoryless model. B. Its expected remaining duration is the original mean minus 10. C. It must end immediately. D. It becomes uniform. 10 pts
6. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. No; averages have no units. B. No; the average has less spread. C. Yes; both are measured in grams. D. Yes; averages always vary more. 10 pts
7. Given that a U(0,12) wait exceeds 4, what shape describes the total wait's conditional density?	A. A point at 4. B. Flat over 0 to 12 unchanged. C. Flat over 4 to 12. D. Normal over all real numbers. 10 pts

Test continues on the other side of the sheet.

MSC 201 **82278**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Uniform model A spans width 4; model B spans width 8. Which density is taller?	A. They have equal height. B. The one with the larger mean regardless of width. C. B. D. A. 10 pts
9. About 99.7% lies within 3 SD in a normal model. About what percent lies outside on both sides combined?	A. 3%. B. 0.15%. C. 99.7%. D. 0.3%. 10 pts
10. All values and their mean and SD are converted from meters to centimeters. What happens to z-scores?	A. They stay unchanged. B. They multiply by 100. C. They all become positive. D. They divide by 100. 10 pts

82278

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. C.

10 pts

4. B.

10 pts

5. A.

10 pts

6. B.

10 pts

7. C.

10 pts

TEST PAGE 2

8. D.

10 pts

9. D.

10 pts

10. A.

10 pts

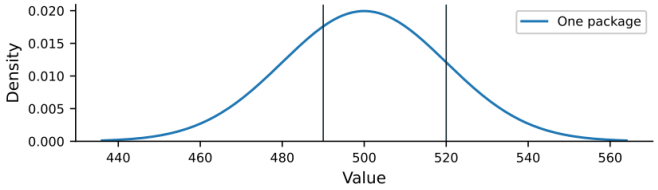
MSC 201 **80269**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A help desk records both the number of requests received in an hour and the exact time until its next request. Which is a continuous variable that an exponential model could describe?	A. Whether there was a request in the hour. B. The exact time until the next request. C. The number of help desks open. D. The number of requests received in the hour. 10 pts
2. A recorded value has z=-1.5. What does that mean?	A. It is above 98.5% of values automatically. B. It has probability -1.5. C. It is 1.5 SD below the mean. D. It is 1.5 original units below zero. 10 pts
3. A package is accepted if 490<mass<520. Which part of its normal curve should be shaded? 	A. Both outside tails. B. Everything above 520. C. Only the point 500. D. Only the area between 490 and 520. 10 pts
4. Durations are rounded to the nearest whole second. Ignoring tie conventions at boundaries, which exact interval reports as 4?	A. 4 to 5 seconds. B. 3.5 to 4.5 seconds. C. Only the exact point 4. D. 0 to 4 seconds. 10 pts
5. P(Z>1.2)=0.1151. By symmetry, what is P(Z<-1.2)?	A. 0.5. B. -0.1151. C. 0.8849. D. 0.1151. 10 pts
6. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. It becomes higher and narrower. B. The center shifts right. C. The total area rises above 1. D. It becomes lower as the curve spreads out. 10 pts

Test continues on the other side of the sheet.

MSC 201 **80269**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. What equation describes a continuous model's 90th percentile k?	A. P(X<=k)=0.90. B. The mean equals 0.90k. C. P(X=k)=0.90. D. P(X>k)=0.90. 10 pts
8. X is uniform from 2 to 10. What can be said about P(X>12)?	A. It is 1/2. B. It is larger than P(X>8). C. It is the density height at 12. D. It is zero. 10 pts
9. A U(0,10) sample has mean 5.4. Must the model be wrong?	A. No; any mean proves the model. B. Yes; all uniform values must equal 5. C. Yes; every sample must average exactly 5. D. No; a sample mean can differ from the model mean. 10 pts
10. An exponential waiting time has mean 5 minutes. What rate per minute matches it?	A. 25. B. 5. C. 0.5. D. 0.2. 10 pts

80269

TEST PAGE 1

1.

B.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

B.

10 pts

5.

D.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

A.

10 pts

8.

D.

10 pts

9.

D.

10 pts

10.

D.

10 pts

MSC 201 **61544**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. How does no arrivals in the next 5 minutes relate to waiting more than 5 minutes for the next arrival?	A. They describe the same event, ignoring a zero-probability boundary. B. The first requires exactly one arrival. C. They are complements. D. They are independent unrelated events. 10 pts
2. P(X>2)=0.50 and P(X>6)=0.125 for an exponential wait. What is P(2<X<6)?	A. 0.125. B. 0.375. C. 0.625. D. 0.50. 10 pts
3. A wait is uniform from 0 to 12 minutes. What is P(3<X<6)?	A. 3/12. B. 6/12. C. 9/12. D. 3/6. 10 pts
4. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. They are equal. B. Both standard deviations are zero. C. The second is larger because its mean is larger. D. The first is larger because it includes zero. 10 pts
5. A normal model has mean 20. What is P(X<20)?	A. The SD. B. 0. C. 0.5. D. 1. 10 pts
6. A value equals its distribution's mean. What is its z-score?	A. The mean. B. The SD. C. 1. D. 0. 10 pts
7. Only 5% exceed cutoff k. Which percentile is k?	A. 50th. B. 100th. C. 5th. D. 95th. 10 pts

Test continues on the other side of the sheet.

MSC 201 **61544**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Under an exact normal model, how do probabilities below 480 and at most 480 compare?	A. Below is always zero. B. They are equal. C. They differ by 1/480. D. At most is larger by the density height. 10 pts
9. A continuous model gives F(6)=0.70. What does that say?	A. 70% of modeled durations are at most 6. B. The density height at 6 is 0.70. C. 70% of durations equal exactly 6. D. The mean is 6. 10 pts
10. A normal model has mean 50 and SD 2. Which interval contains about 99.7%?	A. 46 to 54. B. 48 to 52. C. 44 to 56. D. 50 to 56. 10 pts

61544

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. A.

10 pts

4. A.

10 pts

5. C.

10 pts

6. D.

10 pts

7. D.

10 pts

TEST PAGE 2

8. B.

10 pts

9. A.

10 pts

10. C.

10 pts

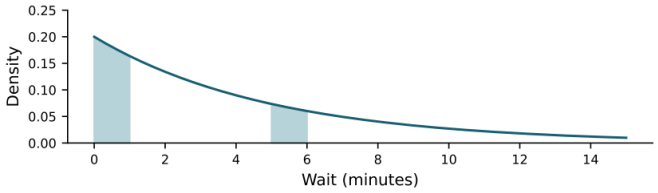
MSC 201 **30015**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. X is uniform from 0 to 10. Which event is more likely?	A. X<6. B. X=6. C. They are equally likely. D. X>6. 10 pts
2. A nonnormal variable has a well-defined mean and SD. Can its z-score automatically be looked up as a standard-normal probability?	A. No; only normal variables can be standardized. B. No; the normal shape assumption still matters. C. Yes; z-scores always imply normality. D. Yes; a positive SD is sufficient. 10 pts
3. For one normal model, P(X<40)=0.20 and P(X<60)=0.85. What is P(40<X<60)?	A. 1.05. B. 0.65. C. 0.20. D. 0.85. 10 pts
4. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It rises by 3. B. It falls by 3. C. It stays unchanged. D. It rises by 90%. 10 pts
5. For an exponential wait, which one-minute interval has more probability? 	A. Both have probability zero. B. They have equal probability. C. 5 to 6 minutes. D. 0 to 1 minute. 10 pts
6. Normal models share SD 4 but have means 10 and 14. Which has more probability below 8?	A. They have equal probabilities. B. The mean-14 model. C. The mean-10 model. D. Both have probability 0.5. 10 pts

Test continues on the other side of the sheet.

MSC 201 **30015**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. The one with more recorded observations regardless of SD. B. They must have equal tails. C. The model with the larger SD. D. The model with the smaller SD. 10 pts
8. Durations are rounded to the nearest whole second. Ignoring tie conventions at boundaries, which exact interval reports as 4?	A. 3.5 to 4.5 seconds. B. Only the exact point 4. C. 4 to 5 seconds. D. 0 to 4 seconds. 10 pts
9. X is uniform from 0 to 12. Given X>4, what is P(X>8)?	A. 8/12. B. 4/12. C. 4/8. D. 8/8. 10 pts
10. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. A constant rate across the whole period. B. Nonnegative waiting times. C. Allowing a long wait. D. Recording waits in minutes. 10 pts

30015

TEST PAGE 1

1. A.

10 pts

2. B.

10 pts

3. B.

10 pts

4. A.

10 pts

5. D.

10 pts

6. C.

10 pts

TEST PAGE 2

7. C.

10 pts

8. A.

10 pts

9. C.

10 pts

10. A.

10 pts

MSC 201 **82175**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. P(Z>1.2)=0.1151. By symmetry, what is P(Z<-1.2)?	A. 0.8849. B. -0.1151. C. 0.1151. D. 0.5. 10 pts
2. X is uniform from 0 to 12. Which symmetric interval contains the middle 50%?	A. 3 to 9. B. 0 to 6. C. 6 to 12. D. 5 to 7. 10 pts
3. A normal model's SD doubles with mean fixed. What happens to its model IQR?	A. It doubles. B. It becomes zero. C. It quadruples. D. It stays unchanged. 10 pts
4. Compare U(0,12) and U(20,32). What is true about their standard deviations?	A. The first is larger because it includes zero. B. The second is larger because its mean is larger. C. Both standard deviations are zero. D. They are equal. 10 pts
5. Normal models have the same mean. For a fixed cutoff below that mean, which has the larger lower-tail probability?	A. They must have equal tails. B. The model with the larger SD. C. The one with more recorded observations regardless of SD. D. The model with the smaller SD. 10 pts
6. Durations are rounded to the nearest whole second. Ignoring tie conventions at boundaries, which exact interval reports as 4?	A. 0 to 4 seconds. B. 4 to 5 seconds. C. Only the exact point 4. D. 3.5 to 4.5 seconds. 10 pts
7. An exponential model has mean wait 4 minutes. What is its SD?	A. 4 minutes. B. 16 minutes. C. 2 minutes. D. 0 minutes. 10 pts

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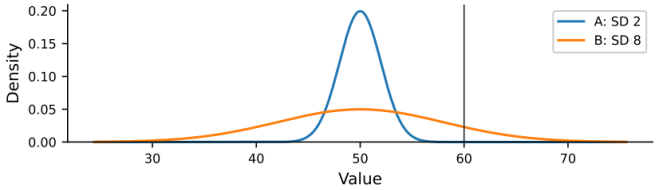
MSC 201 **82175**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. Normal models A and B share mean 50, with SDs 2 and 8. Which has more probability above 60? 	A. Both probabilities are exactly zero. B. B. C. They have equal upper tails. D. A. 10 pts
9. X has mean 50, SD 10. Which standard-normal event corresponds to X<60?	A. Z>1. B. Z<6. C. Z<1. D. Z<-1. 10 pts
10. How does no arrivals in the next 5 minutes relate to waiting more than 5 minutes for the next arrival?	A. They are independent unrelated events. B. The first requires exactly one arrival. C. They are complements. D. They describe the same event, ignoring a zero-probability boundary. 10 pts

82175

TEST PAGE 1

1.

C.

10 pts

2. A.

10 pts

3. A.

10 pts

4.

D.

10 pts

5.

B.

10 pts

6.

D.

10 pts

7. A.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

C.

10 pts

10.

D.

10 pts

MSC 201 **95639**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A probability interval is widened to include its original interval plus a region of positive density. What happens to its probability?	A. It increases. B. It stays unchanged. C. It decreases. D. It exceeds 1 automatically. 10 pts
2. The wait until the next request is modeled as exponential with rate 4 per hour. What is its mean in minutes?	A. 15 minutes. B. 60 minutes. C. 4 minutes. D. 240 minutes. 10 pts
3. A wait is uniform from 0 to 20 minutes. What is its 75th percentile?	A. 75 minutes. B. 15 minutes. C. 20 minutes. D. 5 minutes. 10 pts
4. The SD of a normal model increases while the mean stays fixed. What happens to its peak?	A. The total area rises above 1. B. It becomes lower as the curve spreads out. C. It becomes higher and narrower. D. The center shifts right. 10 pts
5. $P(Z>1.2)=0.1151$. By symmetry, what is $P(Z<-1.2)$?	A. -0.1151. B. 0.5. C. 0.1151. D. 0.8849. 10 pts
6. For a normal model, where is the 10th percentile?	A. At the largest possible value. B. Above the mean. C. At the mean. D. Below the mean. 10 pts
7. X has mean 50, SD 10. Which standard-normal event corresponds to $X<60$?	A. $Z<1$. B. $Z<6$. C. $Z<-1$. D. $Z>1$. 10 pts

Test continues on the other side of the sheet.

MSC 201 **95639**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

8. A U(0,10) sample has mean 5.4. Must the model be wrong?	A. No; a sample mean can differ from the model mean. B. Yes; every sample must average exactly 5. C. Yes; all uniform values must equal 5. D. No; any mean proves the model. 10 pts
9. Arrival rates sharply rise at lunchtime. What is questionable about a single exponential waiting model over the whole day?	A. Nonnegative waiting times. B. Allowing a long wait. C. A constant rate across the whole period. D. Recording waits in minutes. 10 pts
10. A proposed density dips below zero over an interval. What is wrong?	A. It would assign negative area to that interval. B. A density must be flat everywhere. C. Its maximum must equal 1. D. Every valid density must be symmetric. 10 pts

95639

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. B.

10 pts

4. B.

10 pts

5. C.

10 pts

6. D.

10 pts

7. A.

10 pts

TEST PAGE 2

8. A.

10 pts

9. C.

10 pts

10. A.

10 pts

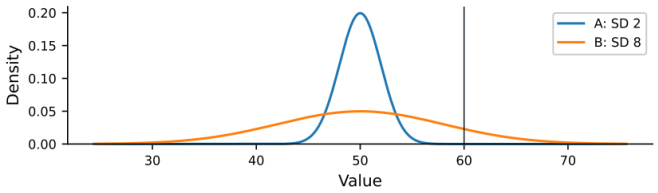
MSC 201 **60980**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. An exact continuous duration has P(X=4)=0. Can a timer report 4 seconds?	A. Only if the density equals zero everywhere. B. Yes; rounding collects a positive-width interval of exact values. C. Only if the model becomes discrete before rounding. D. No; 4 seconds can never be recorded. 10 pts
2. Normal models A and B share mean 50, with SDs 2 and 8. Which has more probability above 60? 	A. Both probabilities are exactly zero. B. B. C. They have equal upper tails. D. A. 10 pts
3. A model describes the mass of one package. Is its SD automatically appropriate for the mean mass of 100 independent packages?	A. No; the average has less spread. B. Yes; both are measured in grams. C. Yes; averages always vary more. D. No; averages have no units. 10 pts
4. For a bell-shaped normal model, about 68% lies within 1 SD. About how much lies outside that central interval?	A. 68%. B. 32%. C. 16% in total. D. 95%. 10 pts
5. The model changes from U(0,10) to U(0,20). What happens to P(X<5)?	A. It becomes 1. B. It halves. C. It stays unchanged. D. It doubles. 10 pts
6. X is uniform from 2 to 10. What can be said about P(X>12)?	A. It is the density height at 12. B. It is larger than P(X>8). C. It is 1/2. D. It is zero. 10 pts

Test continues on the other side of the sheet.

MSC 201 **60980**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

7. Waits are nonnegative. Is that sufficient to establish an exponential model?	A. Yes; every nonnegative variable is exponential. B. Yes; nonnegative variables have equal mean and SD. C. No; the arrival process and distributional assumptions still need support. D. No; exponential values must be negative. 10 pts
8. Compare one exponential duration with the average of many independent durations. What remains the same?	A. The entire density shape. B. The maximum possible duration becomes finite for the average. C. The mean. D. The SD. 10 pts
9. A variable has mean 20 and SD 4. What z-score corresponds to 12?	A. 3. B. -8. C. 2. D. -2. 10 pts
10. A normal model's mean rises by 3 with SD fixed. What happens to its 90th percentile?	A. It falls by 3. B. It rises by 3. C. It rises by 90%. D. It stays unchanged. 10 pts

60980

TEST PAGE 1

1.

B.

10 pts

2.

B.

10 pts

3. A.

10 pts

4.

B.

10 pts

5.

B.

10 pts

6.

D.

10 pts

TEST PAGE 2

7.

C.

10 pts

8.

C.

10 pts

9.

D.

10 pts

10.

B.

10 pts

MSC 201 **30455**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A fitted normal model predicts 2% above a cutoff, while a large observed dataset has 18% above it. What should be investigated?	A. Only the number of decimal places. B. Whether probabilities above 2% are impossible. C. Whether observations should be changed to fit the model. D. Whether the model fits the observed tail. 10 pts
2. Someone reports F(4)=0.6 and F(8)=0.4 for one variable. Why is this impossible?	A. Both numbers are below 1. B. Every cutoff must have the same probability. C. A cumulative probability cannot decrease as the cutoff rises. D. Cumulative functions must always equal 0.5. 10 pts
3. Uniform model A spans width 4; model B spans width 8. Which density is taller?	A. B. B. The one with the larger mean regardless of width. C. They have equal height. D. A. 10 pts
4. A U(0,10) sample has mean 5.4. Must the model be wrong?	A. Yes; every sample must average exactly 5. B. Yes; all uniform values must equal 5. C. No; a sample mean can differ from the model mean. D. No; any mean proves the model. 10 pts
5. The wait until the next request is modeled as exponential with rate 4 per hour. What is its mean in minutes?	A. 60 minutes. B. 15 minutes. C. 240 minutes. D. 4 minutes. 10 pts
6. A value equals its distribution's mean. What is its z-score?	A. The SD. B. 1. C. 0. D. The mean. 10 pts
7. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 90%. B. 10%. C. 5%. D. 45%. 10 pts

Test continues on the other side of the sheet.

MSC 201 **30455**

Unit 3 Test

Name: _____ Date: _____

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Circle one response per question in the right column.

8. An exponential wait has already lasted 10 minutes. Compared with a fresh wait, what is the distribution of the additional wait?	A. It must end immediately. B. The same under the memoryless model. C. Its expected remaining duration is the original mean minus 10. D. It becomes uniform. 10 pts
9. A strongly skewed variable is converted to z-scores. Does it become normal?	A. Yes; subtracting the mean removes the tail. B. No; standardization shifts and rescales without removing skew. C. Yes; every z-score distribution is standard normal. D. No; z-scores cannot be negative. 10 pts
10. For one normal model, $P(X<40)=0.20$ and $P(X<60)=0.85$. What is $P(40<X<60)$?	A. 0.85. B. 0.20. C. 1.05. D. 0.65. 10 pts

30455

TEST PAGE 1

1.

D.

10 pts

2.

C.

10 pts

3.

D.

10 pts

4.

C.

10 pts

5.

B.

10 pts

6.

C.

10 pts

7.

C.

10 pts

TEST PAGE 2

8.

B.

10 pts

9.

B.

10 pts

10.

D.

10 pts

MSC 201 **16096**

Unit 3 Test

Name: _____ Date: _____

Students may use a basic, non-scientific calculator only. No notes or other technology.

Circle one response per question in the right column.

1. A value equals its distribution's mean. What is its z-score?	A. 0. B. 1. C. The mean. D. The SD. 10 pts
2. Durations are rounded to the nearest whole second. Ignoring tie conventions at boundaries, which exact interval reports as 4?	A. 3.5 to 4.5 seconds. B. 4 to 5 seconds. C. Only the exact point 4. D. 0 to 4 seconds. 10 pts
3. After an 11-minute wait, U(0,12) allows at most 1 more minute. An exponential model allows longer additional waits. What explains the difference?	A. Both models are memoryless. B. Exponential waits stop at their mean. C. Uniform waits have a finite endpoint; exponential waits do not. D. Uniform waits are always zero. 10 pts
4. The model changes from U(0,10) to U(0,20). What happens to P(X<5)?	A. It becomes 1. B. It halves. C. It doubles. D. It stays unchanged. 10 pts
5. Waits are nonnegative. Is that sufficient to establish an exponential model?	A. Yes; every nonnegative variable is exponential. B. No; the arrival process and distributional assumptions still need support. C. No; exponential values must be negative. D. Yes; nonnegative variables have equal mean and SD. 10 pts
6. P(X>50)=0.50 and P(X>60)=0.10. Given X>50, what is P(X>60)?	A. 0.10. B. 0.60. C. 0.40. D. 0.20. 10 pts
7. Normal models share SD 4 but have means 10 and 14. Which has more probability below 8?	A. The mean-14 model. B. Both have probability 0.5. C. They have equal probabilities. D. The mean-10 model. 10 pts

Test continues on the other side of the sheet.

MSC 201 **16096**

Unit 3 Test

Name: _____ Date: _____

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Circle one response per question in the right column.

8. A symmetric central normal interval contains 90%. How much area lies in each outside tail?	A. 5%. B. 90%. C. 10%. D. 45%. 10 pts
9. About 95% of a normal model lies within 2 SD. About what percent lies above mean+2 SD?	A. 95%. B. 47.5%. C. 5%. D. 2.5%. 10 pts
10. A uniform wait ranges from 0 to 10. Only 10% exceed a cutoff. Where is that cutoff?	A. At 10. B. At 5. C. At 1. D. At 9. 10 pts

16096

TEST PAGE 1

1. A.

10 pts

2. A.

10 pts

3. C.

10 pts

4. B.

10 pts

5. B.

10 pts

6. D.

10 pts

7. D.

10 pts

TEST PAGE 2

8. A.

10 pts

9. D.

10 pts

10. D.

10 pts